

PRESIDENCY UNIVERSITY

CSE208: Digital Logic Design Laboratory

Lecture 05: Half Subtractor and Full Subtractor

Binary Borrowing, Gate-Level Circuits and Adder–Subtractor Concepts

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Lab 5: What Will You Be Able to Do?

Course outcome CO3: analyze, design and construct combinational circuits.

Understand and design

- Explain **binary borrowing**.
- Build **half** and **full** subtractor truth tables.
- Derive **Difference** and **Borrow** equations.
- Trace borrowing across **multiple bits**.

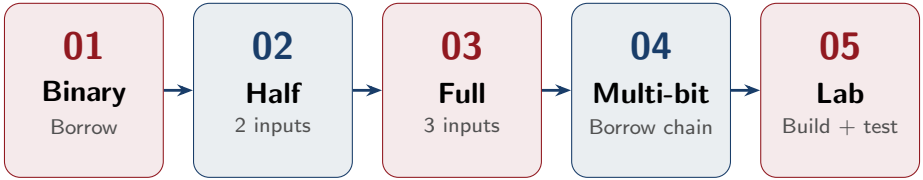
Build and verify

- Construct circuits using **logic ICs**.
- Test **4 + 8** input combinations.
- Compare expected and measured outputs.
- Explain the **combined adder/subtractor** principle.

KEY POINT Required lab activity: **physical construction and truth-table verification**.

Today's Learning Route

Progress from one-bit arithmetic to a working digital circuit.



One question guides the lecture

How does a digital circuit compute $A - B$ and correctly handle a **borrow** from the next binary position?

Binary Subtraction: Why Borrowing Happens

A binary column has values 0 and 1; a borrowed 1 from the next column is worth 2 in the current column.

Single-bit subtraction rules

A	B	Difference	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

When $0 - 1$ occurs

$$\underbrace{0-1}_{\text{original}} \implies \underbrace{2-1}_{\text{after borrowing}} = 1$$

The output difference is **1**, and the circuit signals **Borrow = 1**.

$$\boxed{A - B = D - 2B_{\text{out}}}$$

Example: $0 - 1 = 1 - 2(1) = -1$.

Half Subtractor: Inputs and Outputs

A half subtractor performs one-bit $A - B$ without accepting any previous borrow.



Arithmetic meaning

A is the bit being reduced; B is the bit being subtracted.

Two outputs

D is the one-bit result; B_{out} shows if the next column must supply a borrow.

Half Subtractor: Complete Truth Table

All $2^2 = 4$ input combinations

A	B	D	B_{out}
0	0	0	0
0	1	**1**	**1**
1	0	**1**	0
1	1	0	0

Pattern recognition

- $D = 1$ when inputs **differ**.
- $B_{\text{out}} = 1$ only when $A = 0$, $B = 1$.
- Difference behaves like **XOR**.

KEY POINT Test **every row**; one correct example is not enough to verify the circuit.

Deriving the Half-Subtractor Difference

Read the truth-table rows in which $D = 1$.

Rows producing $D = 1$

A	B	D	Minterm
0	1	**1**	$\bar{A}B$
1	0	**1**	$A\bar{B}$

Sum of products

$$D = \overline{A}B + A\overline{B}$$

$$\boxed{D = A \oplus B}$$

XOR = inputs differ → **Used for Difference**

Deriving the Half-Subtractor Borrow

There is exactly one row where a borrow is needed: $A = 0$, $B = 1$.

Borrow condition

A	B	B_{out}
0	0	0
0	1	**1**
1	0	0
1	1	0

Boolean expression

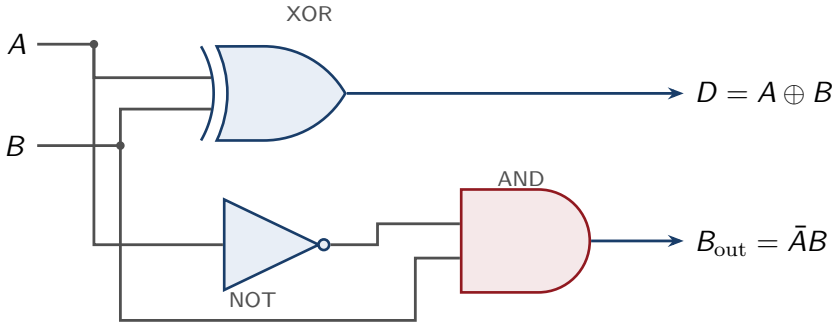
$$\boxed{B_{\text{out}} = \overline{A}B}$$

Invert A, then AND with B.

KEY POINT The same inputs produce two independent functions: $D = A \oplus B$ and $B_{\text{out}} = \bar{A}B$.

Half Subtractor: Gate-Level Circuit

Use one XOR gate, one inverter (NOT), and one AND gate.



KEY POINT XOR produces D; NOT $\rightarrow$ AND produces B_{out}. Crossed wires without a dot are not connected.

Half Subtractor: Worked Output Checks

Case 1: $A = 0,\ B = 1$

$$D = 0 \oplus 1 = \mathbf{1}$$

$$B_{\text{out}} = \bar{0} \cdot 1 = \mathbf{1}$$

$$0 - 1 = 1 - 2(1) = -1$$

A borrow is required

Case 2: $A = 1,\ B = 0$

$$D = 1 \oplus 0 = \mathbf{1}$$

$$B_{\text{out}} = \bar{1} \cdot 0 = \mathbf{0}$$

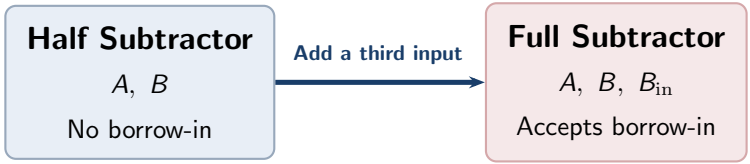
$$1 - 0 = 1 - 2(0) = 1$$

No borrow is required

KEY POINT A difference bit of 1 **does not by itself** tell you whether the subtraction needed a borrow.

Why Is a Half Subtractor Not Enough?

Multi-bit subtraction may inherit a borrow from the immediately less-significant bit.



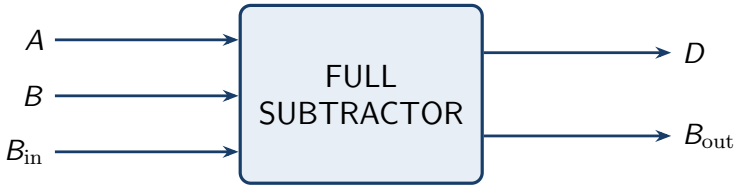
Example

If a column must compute $1 - 0 - 1$, it needs **three inputs**. A half subtractor has only two.

KEY POINT Set $B_{\text{in}} = 0$ to make a full subtractor behave like a half subtractor.

Full Subtractor: Definition and Signals

A full subtractor computes $A - B - B_{\text{in}}$ for one binary position.



KEY POINT The next higher bit receives B_{out} as its incoming borrow.

Full Subtractor: Complete 8-Row Truth Table

All $2^3 = 8$ input combinations

A	B	B_{in}	D	B_{out}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

How to check a row

For $A = 0$, $B = 1$, $B_{\text{in}} = 1$:

$$0 - 1 - 1 = \underbrace{0}_{D} - 2\underbrace{(1)}_{B_{\text{out}}}.$$

Hence $D = 0$ and $B_{\text{out}} = 1$.

Full-Subtractor Difference: XOR of Three Bits

The Difference output is 1 for an odd number of input ones.

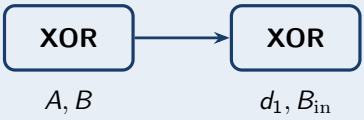
Minterms where $D = 1$

$$D = \bar{A}\bar{B}B_{\text{in}} + \bar{A}B\bar{B_{\text{in}}} + A\bar{B}\bar{B_{\text{in}}} + ABB_{\text{in}}$$

The four rows are 001, 010, 100 and 111.

Use XOR associativity

$$\boxed{D = A \oplus B \oplus B_{\text{in}}}$$



KEY POINT The full-subtractor Difference function is the same three-input XOR as the full-adder Sum function.

Full-Subtractor Borrow: Karnaugh Map

$B_{\text{out}} = 1$ for input rows $001, 010, 011, 111$: $B_{\text{out}} = \Sigma m(1, 2, 3, 7)$.

Three-variable K-map (Gray-code columns)

$A \backslash BB_{\text{in}}$	00	01	11	10
0	0	1	1	1
1	0	0	1	0

Three prime implicants

- $\bar{A}B$
- $\bar{A}B_{\text{in}}$
- BB_{in}

$$B_{\text{out}} = \bar{A}B + \bar{A}B_{\text{in}} + BB_{\text{in}}$$

When Does a Full Subtractor Borrow?

A borrow occurs if the amount to subtract is larger than the available input bit.

$A = 0$ and $(B = 1$ or $B_{\text{in}} = 1)$ **OR** $B = 1$ and $B_{\text{in}} = 1$

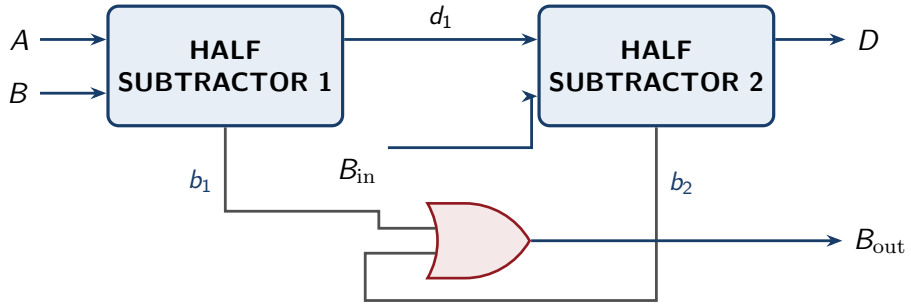
$$B_{\text{out}} = \underbrace{\bar{A}(B + B_{\text{in}})}_{\text{either request when } A=0} + \underbrace{BB_{\text{in}}}_{\text{both requests equal 1}} .$$

Important observation

When $B = B_{\text{in}} = 1$, borrowing is needed **even if** $A = 1$: $1 - 1 - 1 = -1$.

Full Subtractor from Two Half Subtractors

A direct laboratory construction: **2 half subtractors + 1 OR gate**.



At a wire crossing, no junction dot means no electrical connection.

Two-Half-Subtractor Method: Internal Equations

Stage 1: Subtract B from A

$$d_1 = A \oplus B$$

$$b_1 = \bar{A}B$$

Stage 2: Subtract incoming borrow

$$D = d_1 \oplus B_{\text{in}}$$

$$b_2 = \bar{d_1} B_{\text{in}}$$

Combine the two borrows

$$\boxed{B_{\text{out}} = b_1 + b_2}$$

$$= \bar{A}B + \overline{(A \oplus B)} B_{\text{in}}$$

This is equivalent to the K-map expression $\bar{A}B + \bar{A}B_{\text{in}} + BB_{\text{in}}$.

KEY POINT The circuit must OR the internal borrows. It must **not** treat them as a third XOR input.

Two Equivalent Full-Subtractor Realizations

Approach A: Modular construction

$$\boxed{\mathrm{HS}_1 \longrightarrow \mathrm{HS}_2 \longrightarrow \mathrm{OR}}$$

- Reuses **two half subtractors**.
- Intermediate signals: d_1, b_1, b_2.
- Excellent for laboratory tracing.

Approach B: SOP borrow network

$$D = A \oplus B \oplus B_{\mathrm{in}}$$

$$B_{\mathrm{out}} = \bar{A}B + \bar{A}B_{\mathrm{in}} + BB_{\mathrm{in}}$$

- Two XOR gates for Difference.
- NOT/AND/OR network for Borrow.
- Convenient after Boolean minimization.

KEY POINT Both implement the **same truth table**; gate count depends on sharing and IC selection.

Half Subtractor vs Full Subtractor

Comparison at a glance

Feature	Half Subtractor	Full Subtractor
Inputs	A, B	A, B, B_{in}
Difference	$A \oplus B$	$A \oplus B \oplus B_{\text{in}}$
Borrow	$\bar{A}B$	$\bar{A}B + \bar{A}B_{\text{in}} + BB_{\text{in}}$
Truth-table rows	4	8
Borrow-in	Not supported	Supported
Typical use	Bit 0 if initial borrow $= 0$	Higher bits in a borrow chain

Bridge to multi-bit subtraction

A full subtractor can replace a half subtractor by setting $B_{\text{in}} = 0$.

Pause and Predict: Check Your Understanding

Predict without looking at the truth table

For a full subtractor:

1. $A = 1, B = 1, B_{\text{in}} = 1$.
2. $A = 1, B = 0, B_{\text{in}} = 1$.
3. $A = 0, B = 1, B_{\text{in}} = 0$.

For each, determine D and B_{out}.

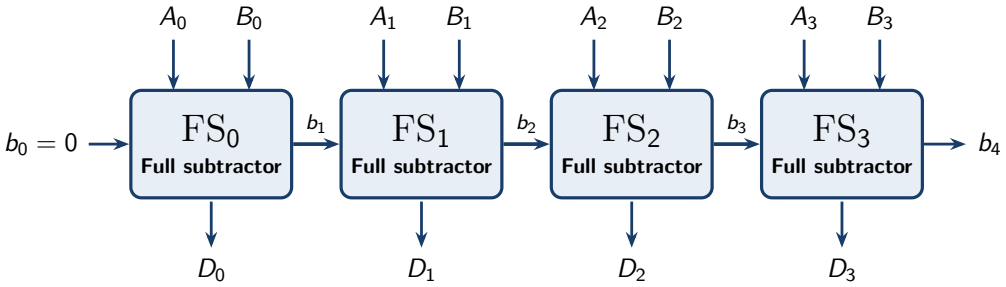
Answer and interpretation

Case	D	B_{out}
1	1	1
2	0	0
3	1	1

Use $A - B - B_{\text{in}} = D - 2B_{\text{out}}$ to check every result.

Building a Multi-Bit Ripple-Borrow Subtractor

Connect the borrow-out of bit i to borrow-in of bit $i+1$. Start at the least-significant bit.



Propagation direction

Borrow flows **from bit 0 to bit 3** (LSB $\rightarrow$ MSB); final b_4 indicates an **unsigned** borrow.

Worked Example: $1001_2 - 0110_2$

Subtract each column from right to left, starting with $b_0 = 0$.

Four full-subtractor stages

Bit	A_i	B_i	b_i	D_i	b_{i+1}
0 (LSB)	1	0	0	**1**	0
1	0	1	0	**1**	**1**
2	0	1	1	0	**1**
3 (MSB)	1	0	1	0	0

Read bits MSB → LSB

$$\begin{array}{r} 1001 \\ -\ 0110 \\ \hline 0011 \end{array}$$

$$9 - 6 = \boxed{3}$$

Final borrow $b_4 = 0$.

KEY POINT Borrow propagates across bit positions even when an intermediate Difference equals 0.

What If the First Number Is Smaller?

A ripple subtractor still produces n Difference bits; final Borrow-out tells you about unsigned underflow.

Example: $0100_2 - 0111_2$

Bit	A_i	B_i	b_i	D_i	b_{i+1}
0	0	1	0	1	1
1	0	1	1	0	1
2	1	1	1	1	1
3	0	0	1	1	1

Interpretation

$$0100_2 - 0111_2 = 1101_2$$

Unsigned: final borrow $= 1$ (underflow).

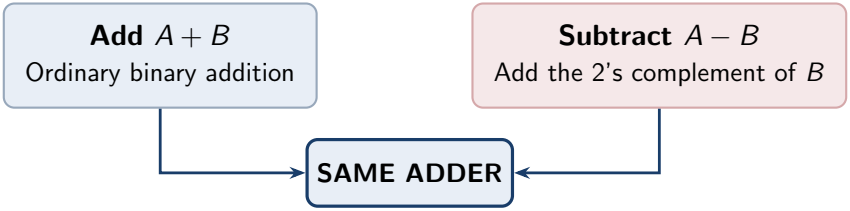
4-bit two's-complement interpretation:

$$1101_2 = -3.$$

KEY POINT Always state the **bit width** and whether the numbers are **signed or unsigned**.

Connecting Subtraction to an Adder

Extension: how a practical arithmetic unit may reuse an adder for both operations.



Core identity for an n-bit circuit

$$A - B \equiv A + (\overline{B}) + 1 \pmod{2^n}$$

The adder operates on n bits; any extra carry is handled separately.

Two's Complement: Invert, Then Add One

For a fixed bit width n, the two's complement of B is $\overline{B} + 1$.



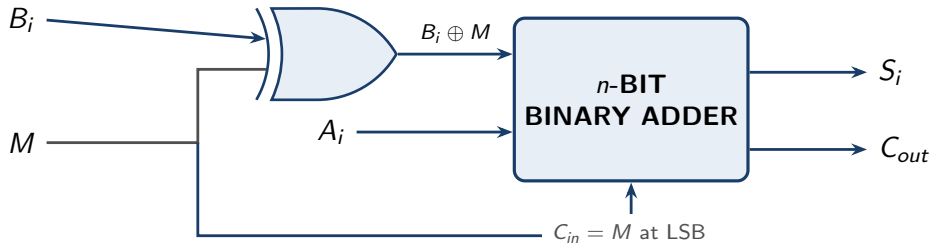
Why it works

For 4 bits, $\overline{0011} = 1100$ and $1100 + 1 = 1101$, which represents -3 in 4-bit two's complement.

KEY POINT Do not omit the **+1**; a bitwise inversion alone is **one's complement**, not two's complement.

Combined Adder/Subtractor: Mode-Controlled Circuit

The mode bit M controls both the inversion of B and the adder's initial carry.



$M = 0$: addition

$B_i \oplus 0 = B_i$, $C_{in} = 0$; result $A + B$.

$M = 1$: subtraction

$B_i \oplus 1 = \bar{B}_i$, $C_{in} = 1$; result $A + \bar{B} + 1$.

Adder/Subtractor Example: $6 - 3$ (4 Bits)

Choose subtraction mode: $M = 1$, so invert B and add the initial carry 1.

Prepare operands

A	0110 (6)
B	0011 (3)
$B \oplus 1111$	1100
Initial C_{in}	1

Perform the addition

$$\begin{array}{r} 0110 \\ 1100 \\ +\ 0001 \\ \hline 1\,0011 \end{array}$$

KEY POINT Result $0011_2 = 3$; carry-out $= 1$ means **no unsigned borrow** for this subtraction.

Adder/Subtractor Example: $3 - 5$ (4 Bits)

An n-bit arithmetic unit retains n result bits; interpreting them needs care.

Prepare operands

A	0011 (3)
B	0101 (5)
$\bar{B}$	1010
Initial carry	1

Perform the addition

$$\begin{array}{r} 0011 \\ 1010 \\ +\ 0001 \\ \hline 0\,1110 \end{array}$$

KEY POINT 1110_2 is -2 in signed 4-bit two's complement. Carry-out $= 0$ indicates an **unsigned borrow**.

Carry, Borrow, and Signed Overflow Are Different

Do not confuse three arithmetic signals

Signal	Meaning
Borrow-out B_{out}	In a ripple subtractor, a 1 means a borrow is needed from the next bit.
Final adder carry C_{out}	When computing unsigned $A + \bar{B} + 1$: $C_{out} = 1$ means **no borrow**; $C_{out} = 0$ means **borrow**.
Signed overflow V	An out-of-range signed answer, tested by $V = C_{\text{in,MSB}} \oplus C_{\text{out,MSB}}$.

Why bit patterns alone are insufficient

$1110_2 = 14$ **unsigned**, but $1110_2 = -2$ in **signed 4-bit two's complement**.

Laboratory: Hardware and Electrical Checklist

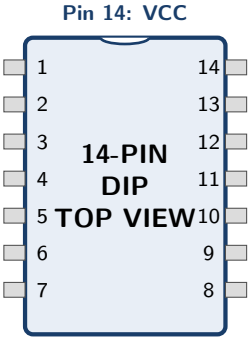
Required materials

- Breadboard, **regulated 5 V** supply.
- **74LS86**: XOR gates.
- **74LS04**: NOT gates.
- **74LS08**: AND gates.
- **74LS32**: OR gates (full subtractor).
- Logic switches, indicators/logic probe, wires.

Before switching power ON

1. Check **VCC = pin 14**, **GND = pin 7**.
2. Match the IC **notch and pin 1**.
3. Use **defined** logic 0/1; do not float inputs.
4. Use suitable LED resistors/buffers; follow IC output-current limits.
5. Turn power **OFF** before rewiring.

Standard 14-Pin DIP: Orientation and Gate Pins



74LS86 / 74LS08 / 74LS32

Gate	Two inputs	Output
1	1, 2	3
2	4, 5	6
3	9, 10	8
4	12, 13	11

74LS04 inverters

Input $\rightarrow$ Output: $1 \rightarrow 2$, $3 \rightarrow 4$, $5 \rightarrow 6$,
$9 \rightarrow 8$, $11 \rightarrow 10$, $13 \rightarrow 12$.

Experiment 1: Wire the Half Subtractor

Aim: implement $D = A \oplus B$ and $B_{\text{out}} = \bar{A}B$; verify four combinations.

Pin-level connection plan

IC	Gate 1 connections
74LS86	XOR: $A \to 1$, $B \to 2$; D from 3
74LS04	NOT: $A \to 1$; $\bar{A}$ from 2
74LS08	AND: $\bar{A} \to 1$, $B \to 2$; B_{out} from 3

Build sequence

1. Power all three ICs correctly.
2. Connect A, B input switches.
3. Complete XOR path to D.
4. Complete NOT $\to$ AND path to B_{out}.
5. Apply 00, 01, 10, 11.

KEY POINT All **four rows** must match the truth table before the circuit is accepted.

Experiment 2: Wire the Full Subtractor

Build two half-subtractor stages and OR their internal borrow signals.

Pin-level wiring (pin 14 = VCC and pin 7 = GND on each IC)

IC	Pins	Input signals $\longrightarrow$ output signal
74LS86 XOR	1,2 $\rightarrow$ 3 4,5 $\rightarrow$ 6	$A, B \longrightarrow d_1$ $d_1, B_{\text{in}} \longrightarrow D$
74LS04 NOT	1 $\rightarrow$ 2 3 $\rightarrow$ 4	$A \longrightarrow \bar{A}$ $d_1 \longrightarrow \bar{d}_1$
74LS08 AND	1,2 $\rightarrow$ 3 4,5 $\rightarrow$ 6	$\bar{A}, B \longrightarrow b_1$ $\bar{d}_1, B_{\text{in}} \longrightarrow b_2$
74LS32 OR	1,2 $\rightarrow$ 3	$b_1, b_2 \longrightarrow B_{\text{out}}$

Verification

Test all **8** patterns $A, B, B_{\text{in}} = 000$ to 111, and record D and B_{out}.

Laboratory Record: Expected vs Observed

Experiment 1: Half subtractor

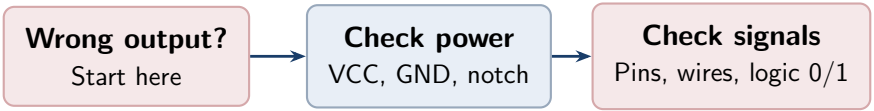
A	B	Expected		Observed	
		D	B_{out}	D	B_{out}
0	0	0	0		
0	1	1	1		
1	0	1	0		
1	1	0	0		

Experiment 2: Full subtractor

A	B	B_{in}	Expected		Observed	
			D	B_{out}	D	B_{out}
0	0	0	0	0		
0	0	1	1	1		
0	1	0	1	1		
0	1	1	0	1		
1	0	0	1	0		
1	0	1	0	0		
1	1	0	0	0		
1	1	1	1	1		

Record actual readings; investigate every mismatch instead of copying expected values.

Troubleshooting: Diagnose Before Rebuilding



Common circuit faults

- Incorrect gate pin or crossed wire.
- Floating or unpowered inputs.
- Incorrect A inverter connection.
- Missing $b_1 + b_2$ OR connection.

Systematic correction

1. Switch **OFF** supply before rewiring.
2. Trace $d_1, \bar{A}, \bar{d}_1, b_1, b_2$.
3. Probe each stage in signal order.
4. Retest the complete truth table.

Quick Viva: Can You Explain Your Circuit?

1. Why does a full subtractor have **three inputs**?
2. When does a half subtractor produce **Borrow = 1**?
3. How do **two half subtractors** produce a full subtractor?
4. Why is the final **Borrow-out** important for unsigned subtraction?
5. If $M = 1$ in a combined adder/subtractor, why are B inverted and the initial carry set to 1?

Challenge

For $A = 0$, $B = 1$, $B_{\text{in}} = 1$, calculate D, B_{out} using **both** equations and arithmetic:
$A - B - B_{\text{in}} = D - 2B_{\text{out}}$.

Lecture 5: What You Should Remember

Half subtractor

$$\boxed{D = A \oplus B}$$

$$\boxed{B_{\text{out}} = \bar{A}B}$$

2 inputs; 4 truth-table rows.

Full subtractor

$$\boxed{D = A \oplus B \oplus B_{\text{in}}}$$

$$\boxed{B_{\text{out}} = \bar{A}B + \bar{A}B_{\text{in}} + BB_{\text{in}}}$$

3 inputs; 8 truth-table rows.

Derive equations | Wire gates | Verify outputs

KEY POINT Next laboratory topic: **Multiplexer** — selecting one of multiple data inputs.

References and Lab Reading

Course-aligned source

- Presidency University, Department of CSE, *CSE208: Digital Logic Design Laboratory Course Outline*, Lab 5: Half Subtractor and Full Subtractor (CO3).

Recommended textbooks from the course outline

1. Mano, M. M. and Ciletti, M. D. (2018). *Digital Design*. 6th ed., Prentice Hall.
2. Tocci, R. J., Widmer, N. S. and Moss, G. L. (2014). *Digital Systems: Principles and Applications*. 11th ed., Pearson.

Hardware reference

Before using a physical 74LS-series IC, confirm its pinout, power limits, and LED/loading requirements against the **datasheet for the exact manufacturer and part**.